Name: Interpretation of Pathways from Spatial Transcriptomic

Purpose:

Spatial transcriptomics makes it possible to study gene expression while preserving information about the spatial organization of tissues. Interpreting these data can be particularly challenging at tissue frontiers, zones where two or more clusters touch, where neighbouring cellular regions meet and different cell populations and biological processes may overlap.

This project aims to support the interpretation of spatial transcriptomics data by integrating information about human genes, biological pathways, and cell types into a unified Knowledge Graph.

The project will focus on the MSigDB Hallmark collection and on human cell-type marker information from CellMarker. By bringing together these two resources, it will be possible to explore the relationships between genes belonging to biological pathways and genes reported as markers of specific cell types. The project will use the HGNC as the soruce for the genes.

In particular, the project will investigate the overlap between pathway-related genes and cell-type marker genes. These overlaps can provide useful biological context when examining pathways identified in different tissue regions or at tissue frontiers. They can help identify cell types whose marker profiles share genes with selected pathways, without assuming that the corresponding pathways are necessarily active in those cell types.

The Knowledge Graph will also support the comparison of activated and suppressed pathways identified by an upstream spatial transcriptomics analysis, allowing the identification of genes shared between the two groups.

The overall goal is to create an integrated and queryable resource that connects biological pathway information with human cell-type annotations and provides additional context for the interpretation of spatial transcriptomics results.

The project aims to address questions like:

- Which human genes are associated with each of the 50 Hallmark biological pathways?
- Which biological pathways contain a given human gene?
- Which cell types have marker genes in common with the genes belonging to a selected biological pathway?
- Which genes are shared between pathways identified as activated and pathways identified as suppressed in a spatial transcriptomics analysis?

Initial Resources:

- Datasets:
  - Molecular Signatures Database (MSigDB), Hallmark gene sets (50 pathways): https://www.gsea-msigdb.org/gsea/msigdb/
  - CellMarker 3.0, human cell-marker information: http://bio-bigdata.hrbmu.edu.cn/CellMarker/
  - HGNC - https://www.genenames.org/
- Ontologies:
  - Gene Ontology (GO) - http://geneontology.org/
  - NCBO BioPortal (Primary Hub for Biomedical Ontologies) - https://bioportal.bioontology.org/

- OBO Foundry (Biomedical Ontology Registry) - https://obofoundry.org/
- Cell Ontology (CL) - https://obofoundry.org/ontology/cl.html
- Human Disease Ontology (DOID) - https://obofoundry.org/ontology/doid.html
- NCI Thesaurus (NCIT - Cell States & Phenotypes) - https://obofoundry.org/ontology/ncit.html
- Pathway Ontology (PW) - https://obofoundry.org/ontology/pw.html
- Gene Ontology Schema (GO) - https://obofoundry.org/ontology/go.html